

Notes: Cull and Xu (JDE, 2003)

Who gets credit? The behavior of bureaucrats and state banks in allocating credit to Chinese state-owned enterprises

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1 Big picture

- **Question.** How did *state banks* and the *government* allocate credit to Chinese state-owned enterprises (SOEs) during 1980–1994? Did banks lend on more commercial grounds than bureaucrats, or were both simply channels of soft-budget-support finance?
- **Setting.** The paper studies Chinese manufacturing SOEs during the reform era, a period in which direct government grants fell sharply and bank lending became a much more important source of finance for fixed investment.
- **Core challenge.** Chinese banks were state owned and operated under important policy-lending constraints, so it is not obvious that bank credit should look much more commercial than direct government transfers. At the same time, SOE reforms changed managerial incentives and may also have generated information that banks could use.
- **Main result.** Bank finance was more strongly associated with profitability than direct government finance. That positive association strengthened during the 1980s, but weakened during the 1990s as bailout responsibilities shifted from the government toward banks.
- **Main signaling result.** Reforms that involved genuine managerial self-selection—especially performance contracts with higher wage elasticity with respect to profits and lower base-profit retention rates—were positively associated with access to bank finance. The same reforms were often negatively related to government transfers.
- **Main interpretation.** Even within a heavily distorted state financial system, bank loan officers had some incentives and some discretion to identify better borrowers. Government bureaucracies, by contrast, used transfers for broader political and social objectives, including support for weak firms.
- **Empirical takeaway.** Direct government transfers were never strongly related to past profitability, while bank credit generally was. But that commercial orientation weakened in the 1990s as banks increasingly became the vehicle for SOE support.
- **Why this paper matters.** The paper is an early and influential micro study of *soft budget constraints*, *state banking*, and *informational frictions* in transition economies. It also shows that reform can matter not only by changing incentives inside firms, but also by producing signals that improve credit allocation.

2 Data and measurement (key objects)

2.1 Observed SOE-level panel

- **Main source.** The paper uses two linked surveys of Chinese state-owned manufacturing enterprises compiled by the Chinese Academy of Social Sciences.
- **1980s panel.** *A Survey of Chinese State Enterprises: 1980–1989* covers 769 SOEs in 21 cities across four provinces: Shanxi, Jilin, Jiangsu, and Sichuan.
- **1990s follow-up.** *A Survey of Chinese SOEs: 1990–1994* revisits more than 600 of the original enterprises; about 680 of the original SOEs remained in the follow-up data.

- **Sample composition.** The sample is manufacturing only, and firms are large: the median SOE in the 1980s survey had 930 employees, the 10th percentile had 304, and the 90th percentile had 3,175.
- **Two-part survey design.** Part one is an accounting table with hundreds of quantitative variables on output, inputs, investment, fixed assets, financing, profit distribution, taxes, and prices. Part two is a manager questionnaire on contracts, autonomy, governance, and reform status.

2.2 Observed financing outcomes

- **Main finance objects.** The paper studies the financing of *fixed investment*.
- **Bank finance.** Defined as either:
 - a dummy for access to any bank finance, or
 - the share of total finance for fixed investment coming from bank loans.
- **Government finance.** Defined as either:
 - a dummy for access to direct government transfers, or
 - the share of total finance for fixed investment coming from direct government grants.
- **Interpretation.** Government finance is treated as pure directed credit. Bank finance is a mixture of policy lending and commercially oriented lending.
- **Other funding sources.** The paper notes that over half of fixed investment was financed by retained funds throughout the sample. So the bank-versus-government comparison is about *how externally allocated funds were distributed*, not about the full financing structure of SOEs.

2.3 Firm characteristics, reforms, and signals

- **Profitability.** Lagged pre-tax profits divided by total value of output.
- **Size and capital intensity.** Log labor and log capital-labor ratio.
- **Payment history.** Unpaid loans relative to total finance, and payment arrears to government relative to value added (available only for the 1990s).
- **Governance.** Dummies for the level of government oversight: central, provincial, city, or county.
- **Signals from performance contracts.** The paper focuses on the Contract Responsibility System and especially:
 - whether the firm had a performance contract,
 - the wage elasticity with respect to profit,
 - the base profit retention rate.
- **Other reform measures.** Autonomy index, share of output under the state plan, discretion over wages, and a new-manager dummy.
- **Market orientation controls.** Export share and markup ratio.

2.4 Unobservables and timing

- **Timing idea.** Lenders allocate funds using information available before the investment is financed, so the paper uses *lagged profitability* as the key performance variable.
- **Structural break.** Because SOE profitability in the sample falls to roughly zero by 1990, the paper allows for a break in the financing process in the 1990s.
- **What is not observed.** The paper does not observe a complete structural lending model, so its strategy is reduced-form: it asks whether bank and government finance respond differently to observable indicators of borrower quality and reform status.

2.5 Why the sample matters

- **Representativeness in finance shares.** The sample tracks the national SOE trend in the shift from direct government grants toward bank finance fairly well.
- **Less profitable than the SOE universe.** The sample SOEs are less profitable than typical SOEs nationally. The authors argue this may make it easier to see the role of soft budget constraints because the sample contains many weak firms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Reduced-form credit-allocation equation

The paper estimates a reduced-form relationship between financing outcomes and firm characteristics:

$$\text{FINANCE}_{it} = \beta_0 + \beta_L L_{it} + \beta_k k_{it} + \beta_c \text{CONTROL}_{it} + \beta_R \text{REFORM}_{it} + \beta_P \text{PAYMENT}_{it} + \delta_p \text{PROFIT}_{i,t-1} + \delta_s \text{SIGNAL}_{it} + \varepsilon_{it}. \quad (1)$$

- **Interpretation.** The paper is modeling the behavior of *fund suppliers*, not firm demand. Given regulated low interest rates, SOE managers are assumed to face effectively elastic demand for external funds. The variation comes from the willingness of banks or bureaucrats to provide finance.
- **Dependent variable.** FINANCE_{it} is alternately a bank-finance variable or a government-finance variable, measured either as access or as share of total finance.
- **Key parameters.** The coefficients of greatest interest are those on lagged profitability and on the signaling variables.

3.2 Why this is a credit-supply regression

- **Regulated interest rates.** Since rates were largely set by the government and often artificially low, the paper treats firms as wanting as much finance as they can get at the prevailing rate.

- **Bank behavior.** Bank loan officers then ration credit across borrowers using whatever information they can gather: past profitability, contracts, payment records, product market indicators, and governance information.
- **Government behavior.** Bureaucrats supplying direct transfers respond to different objectives, including support for distressed SOEs and political or administrative goals.

3.3 Dependent variables and estimators

- **Binary outcomes.** Access to bank finance and access to government finance are estimated with probit.
- **Share outcomes.** Shares of total finance from banks or government are estimated with tobit because the shares are censored at zero and bounded.
- **Why no fixed effects.** The authors argue that access to finance varies too little within firms over time: SOEs that received bank finance in one period tended to receive it throughout, so standard panel fixed-effects methods are not useful here.

3.4 Profitability and bailout responsibilities

To allow the relationship between profitability and finance to vary over time, the paper augments the reduced form with interactions involving year and a 1990s dummy. A schematic version of the main specification is

$$\text{FINANCE}_{it} = \dots + \gamma_1 \text{PROFIT}_{i,t-1} + \gamma_2 (\text{PROFIT}_{i,t-1} \times \text{year}_t) + \gamma_3 (\text{PROFIT}_{i,t-1} \times D_{90}) + \gamma_4 (\text{PROFIT}_{i,t-1} \times D_{90} \times (\text{year}_t - 90)) + \dots \quad (2)$$

- **Interpretation of γ_2 .** If positive, banks become more profitability-sensitive as the 1980s progress.
- **Interpretation of γ_4 .** If negative, the profitability-bank finance link weakens over the 1990s as banks absorb more bailout responsibilities.
- **Economic logic.** This is the heart of the paper: bank lending may become more commercial with reform, but less commercial again if banks are forced to carry more of the soft-budget burden.

3.5 Signals and separation under performance contracts

- **Contract Responsibility System.** Some SOEs sign multi-year performance contracts that link managerial rewards and the enterprise wage bill to firm performance.
- **Incentive intensity versus insurance.** The paper interprets high wage elasticity with respect to profit as a high-powered incentive contract and high base-profit retention as a more insurance-like transfer mechanism.
- **Separation logic.** Efficient firms are more willing to accept contracts with greater performance sensitivity and lower guaranteed transfer. Inefficient firms prefer safer contracts or even no performance contract at all.

- **Empirical implication.** If banks understand this menu, then high wage elasticity and low base retention should predict more bank finance.
- **Time variation.** These signals should become weaker once performance contracts become nearly universal and no longer reflect genuine self-selection.

3.6 Other reforms, controls, and market orientation

- **Autonomy index.** Captures managerial discretion in investment, trade, employment, and asset transactions. The sign is ambiguous because autonomy may either improve performance or be granted to weak firms in need of help.
- **Planned share.** Measures the share of output under the mandatory state plan. The authors initially expect this to reduce bank attractiveness because it limits managerial discretion, though the estimates do not support that view.
- **Discretion over wages.** Could be valuable to banks because it gives managers a tool to align worker effort with firm performance.
- **New manager.** Ambiguous a priori: a new manager may be more entrepreneurial, but also lacks a track record with lenders.
- **Export share and markup ratio.** These proxy international competitiveness and operating margins. Banks should prefer such firms if they are more likely to repay.

3.7 Robustness specifications

- **Profitable-firm subsample.** The paper re-estimates the bank-finance equations on firms with positive profits in at least four of the five years in the 1990s.
- **Working-capital finance.** The paper also studies loans and transfers for working capital in the 1980s to ask whether the same allocation logic applies outside fixed investment.

4 Identification and instruments (what makes the estimates credible)

4.1 What the paper is and is not identifying

- **No excluded instrument.** This paper does not rely on an IV design. Identification is reduced-form and comparative: it asks whether the same observables predict bank finance differently from government finance, and whether those relationships change across the 1980s and 1990s.
- **Not a structural lending model.** The paper does not identify a structural credit supply function with causal reform shocks. Instead, it documents systematic patterns in how funding sources are allocated across firms.

4.2 Why the estimates are still informative

- **Common borrower universe.** Banks and government are allocating funds to the same class of enterprises, in the same environment, over the same period. Differences in coefficients therefore

speak directly to differences in allocation criteria.

- **Lagged profitability.** Using past profitability rather than contemporaneous outcomes helps align the regressors with the information lenders plausibly had when making decisions.
- **Time interactions.** The 1980s-versus-1990s interactions let the paper tie changes in coefficients to the institutional shift from direct government bailouts to bank-mediated support.

4.3 Why the signaling story is plausible

- **Menu of contracts.** Performance contracts differed in incentive intensity and insurance. That is precisely the kind of environment in which a Spence-style separating signal can emerge.
- **Empirical support.** The variables most closely connected to self-selection—wage elasticity and base retention—predict bank finance in the expected direction, especially in the 1980s.
- **Decay of the signal.** The signaling channel weakens as performance contracts become nearly universal, which is exactly what one would expect if their informational value comes from self-selection rather than from the contract label itself.

4.4 Main limitations

- **Endogenous reform adoption.** Firms did not randomly receive reforms. Even with many controls, reform variables may still reflect omitted differences in managerial ability or government treatment.
- **Limited sample.** The sample is manufacturing only, covers four provinces, and is less profitable than the average SOE nationally.
- **No strong within-firm identification.** Financing access changes little within firms, which limits the use of firm fixed effects.
- **Focus on fixed investment.** The main positive results apply to financing of fixed investment; the working-capital evidence is much weaker.

4.5 Relation to the literature

- The paper sits at the intersection of the literature on **soft budget constraints**, **state banking**, and **SOE reform** in China.
- It links firm-reform studies such as Groves et al., Li, and Xu to a banking-allocation question: not just whether reform raises productivity, but whether reform improves the information available to lenders.
- It also brings a **signaling** perspective into transition economics, showing that reform contracts may matter because they reveal information, not only because they change incentives.

5 Tables and figures

5.1 Figure 1: SOE financial structure

Read:

- This figure shows the central background fact of the paper: Chinese SOEs moved away from **direct government transfers** and toward **bank finance** over time.
- In the national data, the bank-loan share of fixed investment rises from 14.3% in 1983 to 24.2% in 1988 and 30.4% in 1992, before easing to 25.7% in 1994. Direct government transfers fall from 40.6% in 1983 to 14.7% in 1988, 6.3% in 1992, and 5.0% in 1994.
- In the sample, bank finance rises from about 15% in 1980 to more than 32% in 1987 and then hovers around 20%–30%. Government finance falls from roughly 25% in the early 1980s to about 2% by 1994.
- This is why the paper studies banks and bureaucrats jointly: the role of the state did not disappear, it was increasingly channeled through banks.

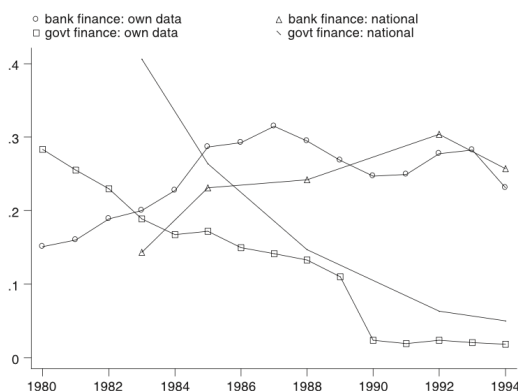


Fig. 1. SOE financial structure. (Source: Sample figures based on the authors' calculations from our data set; national SOE figures are from the *China Statistical Yearbook*.)

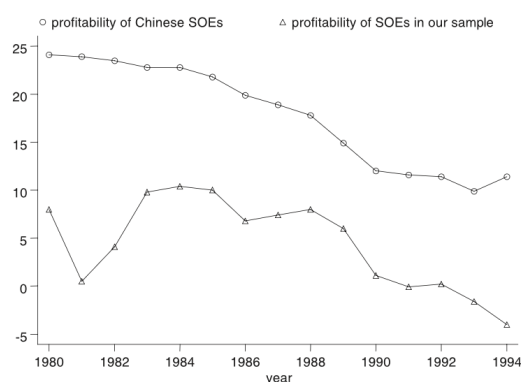


Fig. 2. SOE profitability. Note: National figures come from *Chinese Statistical Yearbook*, various issues. Profitability refers to pre-tax profits divided by the total value of output.

5.2 Figure 2: SOE profitability

Read:

- The figure shows that the sample firms are less profitable than SOEs nationally, but the trend is similar.
- National profitability declines from just under 25% in the early 1980s to roughly 12%–13% by 1992–1994.
- In the sample, profitability falls from about 10%–11% in the mid-1980s to around –4% by 1994.
- This motivates the structural break in the 1990s: by then, many SOEs are weak enough that the politics of bailout become hard to separate from normal lending decisions.

Table 1

Chinese financial system

Bank type	Number branches	Total assets (Renminbi billion)
<i>Central Bank</i>		
People's Bank of China	2529	1758.8
<i>Policy Banks</i>		
State Development Bank of China	—	90.8
Agricultural Development Bank of China	—	406.9
Export-Import Bank of China	—	2.5
<i>State Commercial Banks</i>		
Industrial and Commercial Bank of China	37,039	2633.9
Agricultural Bank of China	63,816	1253.1
Bank of China	12,630	1837.9
People's Construction Bank of China	33,979	1397.5
<i>Nationwide Commercial Banks</i>		
Bank of Communications	1937	305.0
China Trust and Investment Corporation	54	70.7
Investment Bank		
China Everbright Bank	12	20.3
Hua Xia Bank	n.a.	82.9
Min Sheng	—	n.a.
<i>Other Commercial Banks</i>		
China Investment Bank	31	44.5
Guangdong Development Bank	11	43.1
Shenzhen Development Bank	n.a.	15.4
Pudong Development Bank	18	17.8
Shenzhen Merchants Bank	9	44.7
Fujian Industrial Bank	7	19.8
Yantai Housing Saving Bank	5	2.5
Bengbu Housing Saving Bank	n.a.	0.4
<i>Nonbank Financial Institutions</i>		
Rural Credit Cooperatives	50,745	505.3
Urban Credit Cooperatives	5229	214.8
Trust and Investment Companies	391	390.6
People's Insurance Company of China	5240	66.1
China Pacific Insurance Company	n.a.	4.2
Finance Companies	29	27.6
Financial Leasing Companies	11	n.a.
Securities Companies	87	n.a.
Mutual Fund Companies	43	n.a.

“—” is none. “n.a.” means not available. Data are for 1994. Sources: Lardy (1998), *China's Unfinished Economic Revolution*; People's Bank of China, 1995, *China's Financial Outlook*; China Statistical Yearbook 1995; Almanac of Chinese Banking (1995). A similar version of this table is reported in World Bank (1996).

5.3 Table 1: Chinese financial system

Read:

- The table documents the overwhelming dominance of the four state commercial banks in 1994.
- The Industrial and Commercial Bank of China alone has assets of 2,633.9 billion renminbi and more than 37,000 branches; the Agricultural Bank, Bank of China, and People's Construction Bank are also enormous relative to other entrants.
- Policy banks and newer nationwide or regional commercial banks exist, but they are still small compared with the big four.
- So when the paper says “banks,” it is largely talking about a handful of giant state institutions with limited but nonzero commercial discretion.

5.4 Table 2: Variables used in estimation

Read:

- This table is the paper's conceptual roadmap. It tells you what the authors think banks should reward: lagged profitability, signal-bearing reforms, competitiveness, and margins.
- It also shows the contrast the paper is testing: variables that should raise bank finance should often lower government finance if the government is serving a bailout role.

Table 2

Variables used in estimation

Variable names	Description	Available?		Hypothetical link to bank finance
		1980s	1990s	
<i>Dependent variables</i>				
Bank finance	Either share of SOE's total finance that was received from banks or a dummy indicating access to any bank finance.	×	×	
Government finance	Either share of SOE's total finance that was received directly from government or a dummy indicating access to any grants.	×	×	
<i>Explanatory variables</i>				
<i>Firm characteristics</i>				
Profit	Profitability in year $t-1$.	×	×	+
LnCapital ^a	Natural log of capital to labor ratio.	×	×	—
LnLabor	Natural log of the number of workers.	×	×	?
<i>Payment history</i>				
Unpaid loans	Ratio of unpaid loans to total finance.		×	—
Payment arrears	Ratio of unpaid profits, fees, and taxes to the government to total value added.		×	—
<i>Controls (Governance)</i>				
Central, provincial, city, county	Dummy variables indicating the level of government responsible for overseeing the operations of the SOE.	×	×	— for central
<i>Signals</i>				
Performance contract	Dummy variable indicating whether the SOE had a performance contract with the government.	×	×	+
Wage elasticity	Ex ante wage elasticity with respect to profit at the firm level.	×	×	+
Base profit retention	Profit retention rate on 'base profit,' an amount determined by the government.	×		—
<i>Reform variables</i>				
Autonomy index	A principal component index of dummy variables of managerial autonomy in many areas such as investment, foreign trade, employment, and asset transactions.	×	×	+
Planned share	Share of SOE's output under the government's mandatory plan.	×	×	—
Discretion over wages	Dummy variable equal to 1 when managers had discretion to set wages within the SOE.	×	×	+
New manager	Dummy variable equal to 1 if there was a new manager in year t .	×	×	?

Table 2 (continued)

Variable names	Description	Available?		Hypothetical link to bank finance
		1980s	1990s	
<i>Reform variables</i>				
Markup ratio	The ratio of price to marginal cost, as constructed in Li (1997), who used the same data (see the appendices for details).	×	×	+
Export share	Share of total output exported abroad.	×	×	+

^a The two data sets do not allow us to construct a deflator for investment between 1989, the end year of the first data set, and 1990, the first year of the 1990s data set. We thus had to ensure that the results were not sensitive to our method of including $\ln(\text{capital to labor ratio})$ in the *pooled* sample. In the results that we report, the capital to labor ratio for the 1980s is measured in 1989 prices, and that for the 1990s is in 1990 prices. To convert the 1990s observations for capital to 1989 prices, we also include in the regressions $\ln(\text{capital to labor ratio}) \times \text{industry dummy} \times (90s \text{ dummy})$, with the industry dummies at the equivalent of two-digit SIC levels. This in effect allows for an industry-specific deflator for investment between 1989 and 1990. The results do not appear to be sensitive to the way we handled $\ln(\text{capital-labor ratio})$. They remain very similar even without the inclusion of $\ln(\text{capital-labor ratio})$.

- The most important variables are **profitability**, **wage elasticity**, **performance contract**, **base profit retention**, **autonomy index**, **planned share**, **new manager**, **export share**, and **markup ratio**.

5.5 Table 3: Determinants of bank and government financing for fixed investment

Table 3

Determinants of bank and government financing

	Access to bank loans: probit		Share of total finance from banks: tobit		Access to government transfers: probit		Share of total finance from government: tobit	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Observations	4656	4616	4623	4616	4656	4616	4623	4616
Log likelihood ($p > \chi^2$)	−2959.0	−2923.3	−3439.3	−3422.4	−2144.4	−2049.1	−2288.5	−2268.3
Profit _{$t-1$}	0.110 (0.61)	−0.276 (1.20)	−0.012 (0.13)	−0.134 (1.26)	0.323 (1.75)*	0.233 (1.18)	0.093 (1.00)	0.129 (1.26)
Profit _{$t-1$} × year	0.075 (2.44)**	0.239 (3.57)**	0.089 (2.66)**	−0.065 (2.82)**	−0.031 (0.64)	−0.008 (0.87)	−0.008 (0.43)	0.023 (0.67)
Profit _{$t-1$} × 90s dummy	0.089 (0.89)	0.369 (0.89)	0.369 (0.89)	2.384 (0.89)	1.709* (1.33)	2.384 (1.33)	1.200 (1.26)	0.008 (1.26)
Profit _{$t-1$} × 90s dummy × year	−0.589 (2.43)**	−0.290 (2.43)**	−0.290 (2.43)**	−0.290 (2.43)**	−0.348 (0.65)	−0.348 (0.65)	−0.212 (0.78)	−0.008 (0.78)
Wage elasticity	0.140 (2.48)**	0.312 (1.87)*	0.068 (2.45)**	0.150 (3.90)*	−0.078 (1.75)*	−0.300 (0.90)	0.002 (0.04)	−0.148 (1.60)
Wage elasticity × 90s dummy	−0.244 (1.38)	−0.244 (1.38)	−0.244 (1.38)	−0.103 (1.21)	0.300 (1.38)	0.300 (1.38)	0.190 (1.59)	0.000 (1.59)
Performance contract	0.162 (2.91)**	0.210 (1.69)*	0.060 (2.20)**	0.022 (0.37)	−0.076 (0.97)	0.098 (0.77)	−0.074 (1.64)	0.000 (0.78)
Performance contract × 90s dummy	−0.107 (0.76)	−0.107 (0.76)	−0.107 (0.76)	−0.107 (0.76)	−0.357 (1.90)*	−0.357 (1.90)*	−0.212 (2.17)**	−0.212 (2.17)**
Base profit retention	−0.252 (2.54)**	−0.253 (2.54)**	−0.100 (2.03)**	−0.099 (1.99)**	0.182 (1.83)*	0.140 (1.40)	0.119 (2.25)**	0.099 (1.87)*
Discretion over wages	0.064 (1.27)	0.136 (1.70)*	0.027 (1.12)	0.060 (1.62)	−0.068 (1.04)	−0.209 (2.57)**	−0.028 (0.74)	−0.078 (1.79)*
Discretion × 90s dummy	−0.147 (1.41)	−0.147 (1.41)	−0.062 (1.25)	−0.062 (1.25)	0.520 (3.20)**	0.520 (3.20)**	0.240 (2.71)**	0.240 (2.71)**

Autonomy index	−0.150 (4.58)**	−0.136 (1.96)**	−0.071 (4.37)**	−0.077 (2.29)**	−0.063 (1.35)	0.127 (1.84)*	−0.014 (0.51)	0.078 (2.13)**
Autonomy index × 90s dummy	−0.009 (0.11)	−0.009 (0.05)	−0.009 (0.05)	−0.009 (0.05)	−0.009 (0.05)	−0.009 (0.05)	−0.009 (0.05)	−0.009 (0.05)
Planned share	0.156 (2.62)**	0.126 (1.61)	0.081 (2.75)**	0.070 (1.84)*	0.177 (2.47)**	0.231 (2.85)**	0.014 (0.36)	0.013 (0.30)
Planned share × 90s dummy	0.073 (0.61)	0.073 (0.61)	0.073 (0.61)	0.073 (0.61)	0.073 (0.61)	0.073 (0.61)	0.073 (0.61)	0.073 (0.61)
New manager	−0.027 (0.56)	−0.055 (0.89)	−0.025 (1.05)	−0.045 (1.50)	0.205 (3.64)**	0.180 (2.89)**	0.083 (2.75)**	0.071 (2.19)**
New manager × 90s dummy	0.089 (0.43)	0.089 (0.43)	0.089 (0.43)	0.089 (0.43)	0.089 (0.43)	0.089 (0.43)	0.089 (0.43)	0.089 (0.43)
Export share	0.602 (3.77)**	0.636 (3.51)**	0.322 (4.39)**	0.317 (3.84)**	0.129 (0.78)	0.040 (0.23)	0.024 (0.27)	0.008 (0.09)
Export share × 90s dummy	−0.194 (0.49)	−0.194 (0.49)	−0.194 (0.49)	−0.194 (0.49)	−0.194 (0.49)	−0.194 (0.49)	−0.194 (0.49)	−0.194 (0.49)
Markup ratio	0.669 (2.52)**	0.614 (2.29)**	0.354 (2.73)**	0.321 (2.47)**	−0.545 (1.78)*	−0.617 (2.00)**	−0.314 (1.94)*	−0.309 (1.92)*
LnCapital _{$t-1$}	0.015 (0.41)	0.002 (0.06)	−0.018 (0.90)	−0.021 (1.14)	0.117 (2.55)**	0.103 (2.57)**	0.029 (1.35)	0.030 (1.42)
LnLabor _{$t-1$}	0.233 (8.15)**	0.236 (8.16)**	0.073 (5.24)**	0.073 (5.28)**	0.173 (5.44)**	0.177 (5.37)**	0.049 (2.87)**	0.046 (2.67)**
Unpaid loans	−0.009 (0.96)	−0.009 (0.96)	−0.009 (0.96)	−0.009 (0.96)	−0.009 (0.96)	−0.009 (0.96)	−0.009 (0.96)	−0.009 (0.96)
Payment arrears	−0.545 (1.63)	−0.545 (1.63)	−0.545 (1.63)	−0.545 (1.63)	−0.545 (1.63)	−0.545 (1.63)	−0.545 (1.63)	−0.545 (1.63)

*-statistics in parentheses. *, ** and *** represent significance at the 10%, 5% and 1% levels, respectively. Other explanatory variables include year dummies, city dummies, dummies for level of government oversight, $\ln \text{Capital}_{t-1} \times \text{industry dummy} \times (1991-1994 \text{ dummy})$, and missing indicators for base profit retention rates and the planned share.

Read:

- This is the core table. In the bank regressions, the interaction $\text{Profit}_{t-1} \times \text{year}$ is positive and significant, while the additional 1990s slope term is negative and significant. For example, in the access-to-bank-finance probit the coefficients are about 0.075 and -0.589 , and in the bank-share tobit they are about 0.041 and -0.290 .

- The message is clear: bank lending becomes more profitability-sensitive through the 1980s, then less profitability-sensitive in the 1990s as banks inherit more bailout responsibilities.
- Government transfers do *not* show a comparable positive link to profitability. The profit terms in the government regressions are weak and not systematically supportive of commercial screening.
- The signaling variables line up with the theory. Wage elasticity is positively associated with bank finance, while base profit retention is negatively associated with bank finance and positively with government finance. Performance contracts matter more for bank finance in the 1980s and turn strongly negative for government finance in the 1990s.
- Other patterns are also informative: export share and markup ratio are positively associated with bank finance; the autonomy index is negatively associated with bank finance and positively associated with government finance; larger firms receive more of both types of finance; firms under central oversight are more likely to get government finance and less likely to get bank finance.

5.6 Table 4: Profitable-firm subsample

Read:

- This table restricts the sample to firms with positive profits in at least four of the five years in the 1990s.
- The main bank-finance results survive and in some cases become stronger: profitability becomes more strongly related to bank finance in the 1980s, and the signaling coefficients are larger than in the full sample.
- Interestingly, the weakening of the profitability link in the 1990s becomes less visible in this profitable subsample. That is consistent with the idea that the bailout problem mainly concerns weaker firms.
- The new-manager variable turns strongly negative here, suggesting that replacing the manager of an otherwise successful firm creates uncertainty that makes banks more cautious.

5.7 Table 5: Bank and government financing for working capital

Read:

- This table is an important boundary case. The main profitability and signaling patterns are much weaker for **working capital** than for fixed investment.
- Profitability is never significant in the working-capital regressions, and the signaling variables mostly lose their power.
- The interpretation is that working-capital finance was more directly tied to keeping firms operating—meeting payroll and buying intermediate goods—than to screening for high-return investment projects.
- This helps sharpen the main message of the paper: banks looked more commercial than bureaucrats in financing *fixed investment*, but much less so in the financing of day-to-day survival.

Table 4
The determinants of financing for fixed investment: firms with positive profitability at least in 4 of the 5 years in the 1990s

	Access to bank loans: probit		Share of total finance from banks: tobit	
	(1)	(2)	(3)	(4)
Observations	1877	1872	1874	1872
Log likelihood ($p > \chi^2$)	-1115.1	-1105.2	-1284.9	-1273.9
Profit _{t-1}	-0.547 (2.46)**	-0.656 (2.11)**	-0.204 (2.28)**	-0.245 (2.02)**
Profit _{t-1} * year	0.270 (3.67)***	0.327 (2.47)**	0.083 (2.72)***	0.104 (1.95)*
Profit _{t-1} * 90s dummy		-1.826 (0.97)		-0.241 (0.31)
Profit _{t-1} * 90s dummy * year		0.386 (0.66)		0.000 (0.00)
Wage elasticity	0.211 (2.52)**	0.531 (1.95)*	0.064 (1.82)*	0.169 (1.46)
Wage Elasticity * 90s dummy		-0.333 (1.16)		-0.094 (0.78)
Performance contract	-0.005 (0.06)	-0.052 (0.25)	0.003 (0.06)	-0.035 (0.39)
Performance contract * 90s dummy		0.008 (0.03)		0.049 (0.50)
Base profit retention	-0.529 (3.02)***	-0.588 (3.28)***	-0.240 (3.09)***	-0.262 (3.35)***
Discretion over wages	-0.026 (0.31)	-0.098 (0.73)	-0.023 (0.64)	-0.086 (1.53)
Discretion * 90s dummy		0.157 (0.89)		0.131 (1.79)*
Autonomy index	-0.215 (3.72)***	-0.027 (0.26)	-0.093 (3.92)***	-0.038 (0.87)
Autonomy index * 90s dummy		-0.308 (2.51)**		-0.095 (1.83)*
Planned share	0.198 (1.84)*	0.184 (1.38)	0.054 (1.18)	0.047 (0.82)
Planned share * 90s dummy		0.121 (0.55)		0.036 (0.40)
New manager	-0.201 (2.44)**	-0.322 (3.05)***	-0.092 (2.63)***	-0.147 (3.28)***
New manager * 90s dummy		0.347 (2.03)**		0.153 (2.15)**
Export share	0.260 (1.21)	0.227 (0.96)	0.128 (1.42)	0.100 (1.02)
Export share * 90s dummy		0.160 (0.27)		0.152 (0.62)
Markup ratio	0.247 (0.55)	0.105 (0.23)	0.204 (1.02)	0.152 (0.75)
LnCapital _{t-1}	-0.045 (0.65)	-0.052 (0.75)	-0.031 (1.06)	-0.032 (1.11)
LnLabor _{t-1}	0.143 (2.99)***	0.135 (2.79)**	0.014 (0.72)	0.015 (0.76)
Unpaid loans		0.023 (0.85)		-0.002 (0.16)
Payment arrears		-1.024 (0.77)		-0.462 (0.83)

t-statistics in parentheses. *, **, and *** represent significance at the 10%, 5% and 1% levels, respectively. Other explanatory variables include year dummies, city dummies, dummies for level of government oversight, LnCapital_{t-1} × industry dummy × (1991–1994 dummy), and missing indicators for base profit retention rates and the planned share.

Table 5
Determinants of bank and government financing for working capital, 1980s

	Access to bank loans for working capital: probit	Share of bank loans for working capital: tobit	Access to government transfers for working capital: probit	Share of government funds for working capital: tobit
	(1)	(2)	(3)	(4)
Observations	2610	2826	2053	2857
Log likelihood ($p > \chi^2$)	-544.9	289.4	-306.3	1304.1
Profit _{t-1}	-0.326 (1.09)	0.000 (0.01)	0.173 (0.75)	-0.012 (0.66)
Profit _{t-1} * Year	0.103 (1.52)	-0.002 (0.29)	-0.016 (0.25)	0.002 (0.44)
Wage elasticity	0.058 (0.21)	-0.033 (1.23)	-0.773 (1.90)*	-0.005 (0.28)
Performance contract	-0.129 (0.60)	0.041 (2.10)**	0.454 (1.50)	0.003 (0.21)
Base profit retention rate	-0.337 (1.99)**	-0.009 (0.53)	0.043 (0.19)	-0.012 (1.06)
Discretion over wage	0.397 (2.57)**	0.015 (1.18)	0.199 (1.07)	-0.033 (3.61)***
Autonomy index	-0.064 (0.53)	0.017 (1.52)	-0.104 (0.68)	-0.002 (0.20)
Planned share	0.288 (1.98)**	-0.017 (1.25)	0.509 (2.73)***	0.017 (1.75)*
New manager	-0.127 (1.12)	-0.042 (4.20)***	0.045 (0.32)	0.025 (3.46)***
Export share	-0.018 (0.05)	0.083 (2.85)***	0.383 (0.78)	0.065 (3.12)***
Markup ratio	0.764 (1.34)	0.249 (4.84)***	-0.298 (0.29)	-0.011 (0.30)
LnCapital _{t-1}	-0.006 (0.07)	-0.016 (2.55)**	0.363 (4.18)***	0.007 (1.52)
LnLabor _{t-1}	0.325 (5.11)***	0.019 (3.27)***	0.485 (5.64)***	0.033 (7.98)***

t-statistics in parentheses. *, **, and *** represent significance at the 10%, 5% and 1% levels, respectively. Other explanatory variables include year dummies, city dummies, dummies for level of government oversight, and missing indicators for base profit retention rates and the planned share.

6 Short summary

- **Conclusion.** Chinese state banks allocated fixed-investment finance more commercially than direct government bureaucracies, but their commercial orientation weakened when the state shifted bailout responsibilities onto them.
- **Main identification insight.** The paper identifies this by comparing how bank and government finance respond to lagged profitability, reform variables, and their time interactions across the 1980s and 1990s.
- **Main substantive lesson.** Reforms that force managers to self-select can generate useful information for lenders. That informational channel is strongest when the reforms are not yet universal.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that, even in a highly distorted state financial system, banks and bureaucrats did *not* allocate funds in the same way.
- Direct government transfers were not systematically tied to profitability. Bank loans, by contrast, were increasingly tied to profitability during the 1980s.
- That bank-profitability link weakened during the 1990s as banks increasingly became the mechanism through which the state supported weak SOEs.

- Reform variables consistent with managerial self-selection were positively associated with bank finance, suggesting that banks used informational by-products of the reform process.

7.2 Economic intuition

- A bank does not need to be fully private to behave more commercially than a bureaucracy. It only needs some incentive to care about repayment and some discretion over borrower choice.
- Chinese banks had both, albeit imperfectly: employee compensation depended partly on portfolio income, and loan officers had some room to choose among firms within a government credit plan.
- Government bureaus, in contrast, had broader objectives—employment, social stability, regional control, and support of weak firms. So direct transfers naturally looked less profitability-sensitive.
- Performance contracts mattered because they created a noisy but useful signal. Managers willing to expose pay and wages more closely to profits were implicitly telling banks that they expected to perform well.
- Once those contracts became almost universal, the signal became much less informative.

7.3 Takeaway

This paper is best understood as a micro-level study of how soft budget constraints evolve when the state changes the *channel* of support from direct transfers to bank lending. Its key message is not that Chinese banks were efficient in any strong sense. Rather, it is that even constrained, state-owned banks could still do a better job than bureaucracies at directing fixed-investment finance toward more profitable firms, especially when reform created signals that helped distinguish stronger from weaker SOEs. But once banks were asked to absorb more of the bailout burden, that advantage eroded. The broader lesson is that financial reform, enterprise reform, and information revelation are tightly linked in transition economies.